NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

LECTURE 1

The Open Economy: Openness in Goods and Financial Markets

READINGS

1. Olivier Blanchard *Macroeconomics,* 8th Edition (2021), Prentice Hall, Chapter 17.

INTRODUCTION

1. HE2002 Macroeconomics II covers topics in which the economy we looked at is closed i.e. it does not interact with the rest of the world.

2. However, given the degree of integration in the world economy, a crisis that started in one country, can affect nearly every country in the world. For example, the 2008 global financial crisis.

   [Figure 1: Growth in advanced and emerging economies since 2000]

3. Therefore, it is simply not realistic to assume closed economy.

4. We now turn to openness which has three distinct dimensions:

   (a) Openness in goods markets - the ability of consumers and firms to choose between domestic goods and foreign goods.

   (b) Openness in financial markets - the ability of investors to choose between domestic assets and foreign assets.

   (c) Openness in factor markets - the ability of workers to choose where to work and of firms to choose where to locate production.

5. The first choice is governed by the relative price of foreign goods; the second by relative returns on foreign assets.

6. In the short run and in the medium run, openness in factor markets plays much less of role than openness in either goods markets or financial markets, therefore, we shall ignore openness in factor markets.

7. For most countries, open economy considerations have long had substantial effects on economic performance and open economy issues are becoming increasingly important.

8. Lecture 1 focuses on the basic determinants of the trade balance and describes the implications of arbitrage between domestic and foreign bonds.

9. Lecture 2 integrates the trade balance discussion into the closed economy goods market model (the Keynesian cross).

10. Lecture 3 integrates the asset market discussion into the closed economy model of the money market, and develops an open economy IS-LM model.

11. After that, in Lecture 4, we will consider a medium run-model of an open economy with a fixed exchange rate, explore exchange rate crises and the behavior of flexible exchange rates, and discuss the choice of exchange rate regime.

OPENNESS IN GOODS MARKETS Exports and Imports

1. An open economy buys from and sells to the rest of the world. This means that consumers in the open economy are required to choose between domestic goods and foreign goods.

2. As a result, in such decision making, it is important to look at the role of the relative price of domestic goods in terms of foreign goods i.e. the real exchange rate.

3. It should be noted that even countries most committed to free trade have **tariffs** (taxes on imported goods) and **quotas** (restrictions on the quantity of goods that can be imported).

4. Figure 2 shows the evolution of US exports and imports as ratios of GDP from 1960 to 2017.

[Figure 2: US exports and imports as ratios to GDP since 1960]

5. Trade balance, TB, is the difference between exports, X, and imports, M i.e. $TB = X - M$:

   (a) If exports exceed imports $X > M$, there is a **trade surplus** or equivalently, a **positive trade balance**, $TB > 0$,

   (b) If exports is less than imports $X < M$, there is a **trade deficit** or equivalently, a **negative trade balance**, $TB < 0$, and

   (c) If exports equals imports $X = M$, there is a **balance trade**, $TB = 0$.

6. Trade openness is usually measured as:

$$\text{Openness} = \frac{X + M}{Y} \tag{1}$$

   where Y is the total GDP of a country.

7. Another measure of openness is the proportion of aggregate output composed of tradable goods, goods that compete with foreign goods in either domestic markets or foreign markets.

8. Examples of **tradable goods** are cars, computers, pharmaceuticals, etc. Examples of **nontradable goods** are housing, medical services, haircuts, etc.

9. Table 1 shows the ratios of exports to GDP for selected OECD countries in 2017.

[Table 1: Ratios of exports to GDP for selected OECD Countries, 2017]

10. Can exports exceed GDP?

    (a) Since a country cannot export more than it produces, will the export ratio always be less than one?

    (b) No, exports may be larger than GDP because exports and imports may include exports and imports of intermediate goods.

    (c) In 2017, the ratio of exports to GDP in Singapore was 173.

Foreign Exchange Market and Exchange Rates

1. **Foreign exchange market** is a market that involves buying and selling foreign currency.

2. **Nominal exchange rate** is the price of the domestic currency in terms of foreign currency.

    (a) **Nominal appreciation** is an increase in the price of the domestic currency in terms of a foreign currency, i.e., an increase in the exchange rate.

    (b) **Nominal depreciation** is a decrease in the price of the domestic currency in terms of a foreign currency, i.e., a decrease in the exchange rate.

3. Fixed exchange rate system is a system in which two or more countries maintain a constant exchange rate between their currencies.

    (a) **Revaluations** are increases in the exchange rate, and

    (b) **Devaluations** are decreases in the exchange rate.

[Figure 3: The nominal exchange rate between USD and British Pound]

4. **Real exchange rate** is the price of domestic goods relative to foreign goods.

    (a) **Real appreciation** is an increase in the real exchange rate, i.e., an increase in the relative price of domestic goods in terms of foreign goods.

(b) **Real depreciation** is a decrease in the real exchange rate, i.e., a decrease in the relative price of domestic goods in terms of foreign goods.

5. The real exchange rate, the price of U.S. goods in terms of British goods, is

$$\epsilon = \frac{EP}{P^*} \tag{2}$$

where ϵ is the real exchange rate, E is the nominal exchange rate, P and P^* are domestic price and foreign price, respectively.

[Figure 4: The construction of real exchange rate]

[Figure 5: Real and nominal exchange rates between the US and the UK]

6. Bilateral exchange rates are exchange rates between two countries.

7. Multilateral exchange rates say the multilateral real U.S. exchange rate (or U.S. real exchange rate) requires data on the geographic composition of U.S. trade for both exports and imports.

[Figure 6: The U.S. multilateral real exchange rate since 1973]

OPENNESS IN FINANCIAL MARKETS

1. Openness in financial markets allows financial investors to hold both domestic and foreign assets.

2. A country's openness in financial markets has an important implication. It allows the country to run trade surpluses and trade deficits.

The Choice between Domestic and Foreign Assets

1. Openness in financial markets implies that people face a new financial decision: whether to hold domestic assets or foreign assets.

2. Consider the choice between US and UK one-year bonds from the point of view of a US investor:

   (a) Suppose that you have \$1,000 and decide to hold US one-year bonds.

   (b) The interest rate in the US, i_t, is 2%. Then, for \$1,000, you buy US bonds, you will get \$1,020 which is obtained by \$1,000$\times$(1+2%) next year.

   (c) Suppose that you decide to hold UK bonds instead.

   (d) To buy UK bonds, you must first buy pounds. Let the exchange rate, E_t, be \$1 for £0.75. For \$1,000, you will get £750.

3. Let interest rate in UK, i_t^*, be 5%. When next year comes, you will get £750×(1+5%)=£787.5.

4. You will then have to convert your pounds back into dollars.

5. If you expect nominal exchange rate next year,E_{t+1}^e, to be \$1 for £0.78, then each pound will be worth \$(1/0.78)=\$1.28.

6. So, you can expect to have £750×(1+5%)=£787.5 which is £787.5×\$1.28=\$1,010.

7. Now let establish the relationship between interest rates and exchange rates formally.

   [Figure 6: Uncovered interest parity (UIP)]

   (a) Suppose you decide to hold US bonds.

   (b) Let i_t be the one-year US nominal interest rate. Then for every \$1 you put in US bonds, you will get $\$(1 + i_t)$ next year.

   (c) Suppose you decide to hold UK bonds instead.

   (d) To buy UK bonds, you must first buy pounds. Let E_t be the nominal exchange rate between the dollar and pound. For every dollar, you get E_t pounds.

8. Let i_t^* denote the one-year nominal interest rate on UK bonds (in pounds). When next year comes, you will get $E_t(1 + i_t^*)$ pounds.

9. You will then have to convert your pounds back into dollars.

10. If you expect the exchange rate next year to be E_{t+1}^e, then each pound will be worth $(1/E_{t+1}^e)$ dollars.

11. So, you can expect to have $E_t(1 + i_t^*)/E_{t+1}^e$ dollars next year from every dollar you invest now.

12. Assume that you only care about the expected rate of return and hence want to hold only the asset with the highest expected return.

13. In this case, if both UK and US bonds are to be held, they must have the same expected rate of return. This means that:

$$(1 + i_t) = (E_t)(1 + i_t^*)(\frac{1}{E_{t+1}^e}) \tag{3}$$

$$(1 + i_t) = (1 + i_t^*)(\frac{E_t}{E_{t+1}^e}) \tag{4}$$

14. This is called the **uncovered interest parity condition** or simply the **interest parity condition**.

15. The assumption that financial investors will hold only the bonds with the highest expected return is obviously too strong, for two reasons:

(a) Transaction costs are not considered.

(b) Risk is also not taken into consideration.

Interest Rates and Exchange Rates

1. Rewrite E_t/E^e_{t+1} as $1/[1 + (E^e_{t+1} - E_t)/E_t]$ and replacing in Equation (7) gives:

$$(1 + i_t) = \frac{(1 + i^*_t)}{1/[1 + (E^e_{t+1} - E_t)/E_t]} \tag{5}$$

2. This gives a relation between domestic nominal interest rate, i_t, the foreign interest rate, i^*_t, and the expected rate of appreciation of domestic currency, $(E^e_{t+1} - E_t)/E_t$.

3. As long as interest rates or the expected rate of appreciation are not too large, then, a good approximation implies:

$$i_t = i^*_t - \frac{E^e_{t+1} - E_t}{E_t} \tag{6}$$

4. This means that the domestic interest rate must be equal to foreign interest rate minus the expected rate of appreciation of the domestic currency.

5. Or equivalently, the domestic interest rate must be equal the foreign interest rate minus the expected rate of depreciation of the foreign currency.

The Balance of Payments

1. A country's international transactions are recorded in the **balance-of-payments**, BP.
2. BP has three main components:
   (a) Current account, CA
   (b) Financial account, FA, and
   (c) Capital account, KA.
3. Therefore, BP is usually written as:
$$BP = CA + FA + KA \tag{7}$$
4. Since capital account balance is usually negligible, we are not discussing it here. Hence,
$$BP = CA + FA \tag{8}$$
5. Any transaction resulting in a payment to foreigners is entered in the BoP accounts as a debit and is given a negative $(-)$ sign. Any transaction resulting in a receipt from foreigners is entered as a credit and is given a positive $(+)$ sign.
6. The current account records exports and imports of goods and services and international receipts or payments of income.
7. Exports and income receipts enter as positive. Imports and income payments enter as negative.
8. The financial account keeps records of sales of assets to foreigners (financial inflows, FI) and purchases of assets located abroad by domestic residents (financial outflows, FO).
9. Sales of assets to foreigners enter as positive. Purchases of assets located abroad enter as negative.

[Table 2: The US balance of payments in 2018 in billions of USD]

10. BoP follows a fundamental accounting principle known as *double-entry bookkeeping* with each transactions enters the BoP twice, once with a positive sign and another with a negative sign.
11. Because any international transactions automatically gives rise to two offsetting entries in the BoP, we have
$$\text{Current account balance} = -\text{Financial account balance} \tag{9}$$

12. The current account, on the other hand, consists of three components:

$$\text{Current account } = \text{Trade balance} + \text{Income balance} + \text{Net unilateral transfer} \tag{10}$$

13. The trade balance or balance on goods and services is the difference between exports and imports of goods and services. It consists of

    (a) Merchandise trade balance or balance on goods, and

    (b) Services balance.

14. The merchansise trade balance keeps record of net exports of goods.

15. The services balance keeps record of net receipts from items such as transportation, travel expenditures and legal assistance.

16. The income balance has

    (a) Net investment income, and

    (b) Net investment compensation to employees.

17. Net investment income is the difference between income receipts on US-owned assets abroad and income payments on foreign-owned assets in the US. It includes items such as international interest, dividend payments and earnings of domestically owned firms operating abroad.

18. Net investment compensation to employees includes

    (a) As positive entries: (1) earnings of US residents employed temporarily abroad, (2) earnings of US residents employed by foreign governments in US, and (3) earnings of US residents employed by international organizations in US.

    (b) As negative entries: (1) foreign workers who commute to work in US, (2) foreign students studying in US, (3) foreign professionals temporarily residing in US, and (4) foreign temporary workers in US.

19. Net unilateral transfers keeps record of the difference between gifts. One common item is private remittances.

20. The financial account has two main components as follows:

$$\begin{aligned} \text{Financia account} \quad = \quad & \text{Increased in foreign-owned assets in the US} && (11) \\ & -\text{Increased in US-owned assets abroad} && (12) \end{aligned}$$

21. Note that foreign-owned assets in the US includes

   (a) US securities held by foreign residents
   (b) US currency held by foreign residents
   (c) US borrowing from foreign banks
   (d) Foreign direct investment in the US

22. US-owned assets abroad are

   (a) Foreign securities
   (b) US bank lending to foreigners
   (c) US direct investment abroad

23. Note that when US borrows \$1 from foreigners, it is selling them an asset - a promise that tey will be repaid \$1, with interest, in the future. Such transactions enters the financial account with a positive sign because the loan is itself a payment to the US or a financial inflow (also sometimes called a capital inflow).

24. When US lends abroad, however, a payment is made to foreigners. This transaction involves the purchase of an asset from foreigners and is called a financial outflow (or, alternatively, a capital outflow).

25. There is another type of transaction that merits separate discussion. This type of transaction is the purchase or sale of official reserve assets by central banks.

26. Official international reserves are foreign assets held by central banks as a cushion against national economic misfortune. At one time, official reserves consisted largely gold, but today central bank's reserves include substantial foreign financial assets.

27. Central banks often buy or sell international reserves in private asset markets to affect macroeconomic conditions in their economies. Official transactions of this type are called **official foreign exchange interventions**.